

1.Creación de usuario admin y usuarios sog1 y sog2

```
root@instance-20260707-1651:/home/ubuntu/johan_baracaldo# adduser adminn
Adding user `adminn' ...
Adding new group `adminn' (1004) ...
Adding new user `adminn' (1004) with group `adminn' ...
Creating home directory `/home/adminn' ...
Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for adminn
Enter the new value, or press ENTER for the default
    Full Name []: admin
    Room Number []:
    Work Phone []:
    Home Phone []:
    Other []:
Is the information correct? [Y/n] y
root@instance-20260707-1651:/home/ubuntu/johan_baracaldo# echo Johan David Baracaldo Caicedo
Johan David Baracaldo Caicedo
root@instance-20260707-1651:/home/ubuntu/johan_baracaldo#
```

```
root@instance-20260707-1651:/home/ubuntu# cd johan_baracaldo/
root@instance-20260707-1651:/home/ubuntu/johan_baracaldo# adduser admin
adduser: The group `admin' already exists.
root@instance-20260707-1651:/home/ubuntu/johan_baracaldo# adduser sog1
Adding user `sog1' ...
Adding new group `sog1' (1002) ...
Adding new user `sog1' (1002) with group `sog1' ...
Creating home directory `/home/sog1' ...
Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for sog1
Enter the new value, or press ENTER for the default
    Full Name []: sog1
    Room Number []: 1
    Work Phone []:
    Home Phone []:
    Other []:
Is the information correct? [Y/n] y
root@instance-20260707-1651:/home/ubuntu/johan_baracaldo# adduser sog2
Adding user `sog2' ...
Adding new group `sog2' (1003) ...
Adding new user `sog2' (1003) with group `sog2' ...
Creating home directory `/home/sog2' ...
Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for sog2
Enter the new value, or press ENTER for the default
    Full Name []: sog2
    Room Number []: 2
    Work Phone []:
    Home Phone []:
    Other []:
Is the information correct? [Y/n] y
root@instance-20260707-1651:/home/ubuntu/johan_baracaldo# echo Johan David Baracaldo Caicedo
Johan David Baracaldo Caicedo
root@instance-20260707-1651:/home/ubuntu/johan_baracaldo#
```

2. Creación de proceso .sh para apertura de Wikipedia cada 40 segundos y asignación al usuario sog1

```
root@instance-20260707-1651:/home/ubuntu/johan_baracaldo# cat > wikipedia.sh
#!/bin/bash
# Proceso disparado para g1 - abre Wikipedia cada 40 segundos

while true; do
    # Abre una página aleatoria de Wikipedia
    xdg-open "https://es.wikipedia.org/wiki/Special:Random" &
    echo "[$(date)] Abriendo Wikipedia - PID del navegador: $!"
    sleep 40
done

^C
root@instance-20260707-1651:/home/ubuntu/johan_baracaldo# chmod +x wikipedia.sh
root@instance-20260707-1651:/home/ubuntu/johan_baracaldo# chown sog1:sog1 wikipedia.sh
root@instance-20260707-1651:/home/ubuntu/johan_baracaldo# mv wikipedia.sh /home/sog1/
root@instance-20260707-1651:/home/ubuntu/johan_baracaldo# echo Johan David Baracaldo Caicedo
Johan David Baracaldo Caicedo
root@instance-20260707-1651:/home/ubuntu/johan_baracaldo# |
```

```
root@instance-20260707-1651:/home/ubuntu/johan_baracaldo# sudo -u sog1 bash -c "cd /home/sog1 && nohup ./wikipedia.sh > wiki.log 2>&1 &"
root@instance-20260707-1651:/home/ubuntu/johan_baracaldo# ps -u sog1 -o pid,user,cmd
  PID USER      CMD
 174201 sog1    bash -c cd /home/sog1 && nohup ./wikipedia.sh > wiki.log 2>&1 &
 174202 sog1    /bin/bash ./wikipedia.sh
 174205 sog1    sleep 40
root@instance-20260707-1651:/home/ubuntu/johan_baracaldo# echo Johan David Baracaldo Caicedo
Johan David Baracaldo Caicedo
root@instance-20260707-1651:/home/ubuntu/johan_baracaldo# |
```

3. Creación del proceso .sh para sumas infinitas y asignación al usuario sog2

```
root@instance-20260707-1651:/home/ubuntu/johan_baracaldo# cat > sumas_infinitas.sh
#!/bin/bash
# Proceso de sumas para g2 - suma hasta que pasen 40 segundos

contador=0
suma_total=0
tiempo_fin=$(( $(date +%s) + 40 ))

echo "Iniciando sumas... el proceso durará 40 segundos"

while [ $(date +%s) -lt $tiempo_fin ]; do
    suma_total=$((suma_total + contador))
    echo "[${contador}] Suma acumulada: $suma_total - $(date +%T)"
    contador=$((contador + 1))
    sleep 1
done

echo "Finalizado. Total sumado: $suma_total"

^C
root@instance-20260707-1651:/home/ubuntu/johan_baracaldo# chmod +x sumas_infinitas.sh
root@instance-20260707-1651:/home/ubuntu/johan_baracaldo# chown sog2:sog2 sumas_infinitas.sh
root@instance-20260707-1651:/home/ubuntu/johan_baracaldo# mv sumas_infinitas.sh /home/sog2/
root@instance-20260707-1651:/home/ubuntu/johan_baracaldo# sudo -u sog2 bash -c "cd /home/sog2 && nohup ./sumas_infinitas.sh > sumas.log 2>&1 &"
root@instance-20260707-1651:/home/ubuntu/johan_baracaldo# ps -u sog2 -o pid,user,cmd
  PID USER      CMD
 174258 sog2    bash -c cd /home/sog2 && nohup ./sumas_infinitas.sh > sumas.log 2>&1 &
 174259 sog2    /bin/bash ./sumas_infinitas.sh
 174323 sog2    sleep 1
root@instance-20260707-1651:/home/ubuntu/johan_baracaldo# echo Johan David Baracaldo Caicedo
Johan David Baracaldo Caicedo
root@instance-20260707-1651:/home/ubuntu/johan_baracaldo#
```

```

root@instance-20260707-1651:/home/ubuntu/johan_baracaldo# usermod -aG sudo adminn
root@instance-20260707-1651:/home/ubuntu/johan_baracaldo# groups adminn
adminn : adminn sudo
root@instance-20260707-1651:/home/ubuntu/johan_baracaldo# sudo gpasswd -d sog1 sudo
Removing user sog1 from group sudo
gpasswd: user 'sog1' is not a member of 'sudo'
root@instance-20260707-1651:/home/ubuntu/johan_baracaldo# sudo gpasswd -d sog2 sudo
Removing user sog2 from group sudo
gpasswd: user 'sog2' is not a member of 'sudo'
root@instance-20260707-1651:/home/ubuntu/johan_baracaldo# sudo usermod -G sog1 sog1
root@instance-20260707-1651:/home/ubuntu/johan_baracaldo# sudo usermod -G sog2 sog2
root@instance-20260707-1651:/home/ubuntu/johan_baracaldo# groups sog1
sog1 : sog1
root@instance-20260707-1651:/home/ubuntu/johan_baracaldo# groups sog2
sog2 : sog2
root@instance-20260707-1651:/home/ubuntu/johan_baracaldo# echo Johan David Baracaldo Caicedo
Johan David Baracaldo Caicedo
root@instance-20260707-1651:/home/ubuntu/johan_baracaldo#

```

3. Ingreso al usuario administrador para validar los proceso corriendo en sog1 y sog2 para después usar kill y eliminar dichos procesos

```

adminn@instance-20260707-1651:~$ ps -u sog1 -o pid,user,cmd
  PID USER      CMD
  174201 sog1    bash -c cd /home/sog1 && nohup ./wikipedia.sh > wiki.log 2>&1 &
  174202 sog1    /bin/bash ./wikipedia.sh
  174539 sog1    sleep 40
adminn@instance-20260707-1651:~$ ps -u sog2 -o pid,user,cmd
  PID USER      CMD
  174545 sog2    bash -c cd /home/sog2 && nohup ./sumas_infinitas.sh > sumas.log 2>&1 &
  174546 sog2    /bin/bash ./sumas_infinitas.sh
  174626 sog2    sleep 1
adminn@instance-20260707-1651:~$ sudo kill -9 PID_DE_SOG1
[sudo] password for adminn:
kill: failed to parse argument: 'PID_DE_SOG1'
adminn@instance-20260707-1651:~$ sudo pkill -u sog1 -f wikipedia
adminn@instance-20260707-1651:~$ sudo pkill -u sog2 -f sumas
adminn@instance-20260707-1651:~$ ps -u sog1
  PID TTY          TIME CMD
  174697 ?            00:00:00 sleep
adminn@instance-20260707-1651:~$ ps -u sog2
  PID TTY          TIME CMD
adminn@instance-20260707-1651:~$ echo Johan David Baracaldo Caicedo
Johan David Baracaldo Caicedo
adminn@instance-20260707-1651:~$

```

4. Intento de eliminación de procesos con sog2 el cual carece de permisos de administrador por lo cual el sistema no lo permite, para este paso se ponen a correr nuevamente los procesos de Wikipedia y suma infinita para hacer la prueba desde sog2

```
root@instance-20260707-1651:/home/ubuntu/johan_baracaldo# sudo -u sog2 bash -c "cd /home/sog2 && nohup
./sumas_infinitas.sh > sumas.log 2>&1 &"
root@instance-20260707-1651:/home/ubuntu/johan_baracaldo# sudo -u sog1 bash -c "cd /home/sog1 && nohup
./wikipedia.sh > wiki.log 2>&1 &"
root@instance-20260707-1651:/home/ubuntu/johan_baracaldo# su - sog2
sog2@instance-20260707-1651:~$ ps -u sog1
  PID TTY          TIME CMD
 174916 ?            00:00:00 bash
 174917 ?            00:00:00 wikipedia.sh
 174920 ?            00:00:00 sleep
sog2@instance-20260707-1651:~$ kill PID_DE_SOG1
-bash: kill: PID_DE_SOG1: arguments must be process or job IDs
sog2@instance-20260707-1651:~$ ps -u sog1
  PID TTY          TIME CMD
 174916 ?            00:00:00 bash
 174917 ?            00:00:00 wikipedia.sh
 174920 ?            00:00:00 sleep
sog2@instance-20260707-1651:~$ sudo kill PID_DE_SOG1
[sudo] password for sog2:
sog2 is not in the sudoers file.  This incident will be reported.
sog2@instance-20260707-1651:~$ ps -u sog1
  PID TTY          TIME CMD
 174916 ?            00:00:00 bash
 174917 ?            00:00:00 wikipedia.sh
 175012 ?            00:00:00 sleep
sog2@instance-20260707-1651:~$
```